Chapter 4

Functors and Natural Transformations

4.1 Functors

Definition 4.1.1 (Functor)**.** A *functor* $\boldsymbol{F}$ between categories $\mathcal{C}$ and $\mathcal{D}$ (notation $\boldsymbol{F} : \mathcal{C} \longrightarrow \mathcal{D}$) is specified by:

- an operation assigning to each object A in $\mathcal{C}$ an object $\boldsymbol{F}A$ in $\mathcal{D}$
- an operation assigning to each morphism $f : A \longrightarrow B$ in $\mathcal{C}$ a morphism $\boldsymbol{F}f : \boldsymbol{F}A \longrightarrow \boldsymbol{F}B$ in $\mathcal{D}$ which satisfies the following:

$$
\begin{aligned}
\boldsymbol{F}(id_A) &= id_{\boldsymbol{F}A} \\
\boldsymbol{F}(g \circ f) &= \boldsymbol{F}g \circ \boldsymbol{F}f
\end{aligned}
$$

i.e. the following diagram commutes.

$$
\begin{tikzcd}
\mathcal{C}: & A \arrow[loop left, "id_A"] \arrow[r, "f"] \arrow[d, dashed, "\boldsymbol{F}"] & B \arrow[r, "g"] \arrow[d, dashed, "\boldsymbol{F}"] & C \arrow[d, dashed, "\boldsymbol{F}"] \\
\mathcal{D}: & \boldsymbol{F}A \arrow[loop left, "id_{\boldsymbol{F}A}"'] \arrow[r, "\boldsymbol{F}f"] & \boldsymbol{F}B \arrow[r, "\boldsymbol{F}g"] & \boldsymbol{F}C
\end{tikzcd}
$$

Remark 4.1.2. By a functor $\boldsymbol{F} : \mathcal{A} \longrightarrow \mathcal{B}$, any commutative diagram in $\mathcal{A}$ is sent to a commutative diagram in $\mathcal{B}$.

Example 4.1.3 (Powerset functor)**.** The *powerset functor* $\mathcal{P}ow(\_) : \mathcal{S}et \longrightarrow \mathcal{S}et$ sends each object X to $\mathcal{P}ow(X)$ and each arrow $f : X \longrightarrow Y$ to $X' \mapsto \{f(x) \mid x \in X'\}$.

Example 4.1.4 (Product functor)**.** Suppose that a category $\mathcal{C}$ has binary products. The *product functor* $\_ \times A : \mathcal{C} \longrightarrow \mathcal{C}$ sends each object B to $B \times A$ and each arrow $f : B \longrightarrow C$ to $f \times id_A$.

Example 4.1.5 (Exponentiation functor)**.** Suppose that a category $\mathcal{C}$ is Cartesian closed. The *exponentiation functor* $\_^A : \mathcal{C} \longrightarrow \mathcal{C}$ sends each object B to B^A and each arrow $f : B \longrightarrow C$ to $f^A : B^A \longrightarrow C^A$.

Example 4.1.6 (Homomorphism)**.** The *homomorphisms* between posets are functors.

Exercise 4.1.7. Confirm that each of the examples above actually conforms to the definition of functor.

Example 4.1.8. The functor $\boldsymbol{F} : \mathcal{S}et \longrightarrow \mathcal{M}on$ takes a set to its Kleene closure, which sends each set A to A^*, and each set function f to $map(f)$, where $map(f) : A^* \longrightarrow B^*$ is defined by:

$$map(f)([a_1, \ldots, a_n]) \quad \stackrel{def}{\equiv} \quad [f(a_1), \ldots, f(a_n)]$$

where $[a_1, \ldots, a_n]$ is any element of A^*. To verify that $\boldsymbol{F}$ is indeed a functor, check:

$$\begin{aligned}
\boldsymbol{F}(id_A)([a_1, \ldots, a_n]) &\stackrel{def}{\equiv} map(id_A)([a_1, \ldots, a_n]) \\
&= id_{A^*}([a_1, \ldots, a_n]) \\
&\stackrel{def}{\equiv} id_{\boldsymbol{F}A}([a_1, \ldots, a_n])
\end{aligned}$$

$$\begin{aligned}
\boldsymbol{F}(gf)([a_1, \ldots, a_n]) &\stackrel{def}{\equiv} map(gf)([a_1, \ldots, a_n]) \\
&= [gf(a_1), \ldots, gf(a_n)] \\
&= map(g)([f(a_1), \ldots, f(a_n)]) \\
&= map(g)(map(f)([a_1, \ldots, a_n])) \\
&= \boldsymbol{F}g \circ \boldsymbol{F}f([a_1, \ldots, a_n])
\end{aligned}$$

Exercise 4.1.9. $\mathcal{C}at$ is a category of categories with categories as objects and functors as arrows.

1. For each category $\mathcal{A}$, define a functor $\boldsymbol{id_{\mathcal{A}}}$ as an identity functor for $\mathcal{A}$.

2. Show that functors are composable and the composition is unitary with the identity functor above, and associative.

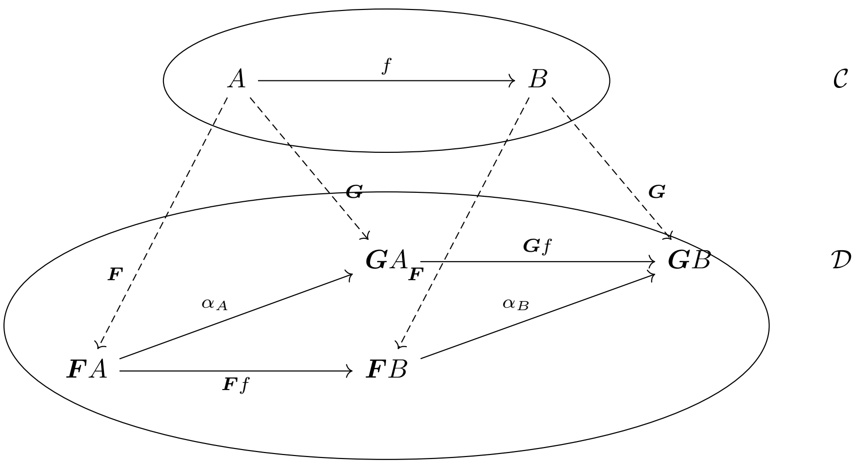
4.2 Natural Transformations

Definition 4.2.1 (Natural transformation)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $\boldsymbol{F}, \boldsymbol{G} : \mathcal{C} \longrightarrow \mathcal{D}$ be functors. A *natural transformation* α from $\boldsymbol{F}$ to $\boldsymbol{G}$ (notation $\alpha : \boldsymbol{F} \stackrel{\centerdot}{\longrightarrow} \boldsymbol{G}$) is specified by:

- an operation assigning to each object A in $\mathcal{C}$ a morphism $\alpha_A : \boldsymbol{F}A \longrightarrow \boldsymbol{G}A$ in $\mathcal{D}$, such that for any morphism $f : A \longrightarrow B$ in $\mathcal{C}$, we have:

$$\boldsymbol{G}f \circ \alpha_A = \alpha_B \circ \boldsymbol{F}f$$

i.e. the following diagram commutes.



Example 4.2.2. Given the functor $\boldsymbol{F} : \mathcal{S}et \longrightarrow \mathcal{M}on$, we can define a natural transformation $rev : \boldsymbol{F} \stackrel{\centerdot}{\longrightarrow} \boldsymbol{F}$ which has components $rev_A : A^* \to A^*$ defined

by:

$$rev_A([a_1, \ldots, a_n]) \stackrel{def}{\equiv} [a_n, \ldots, a_1]$$

where A is a set and $[a_1, \ldots, a_n] \in A^*$. To verify that this is indeed a natural transformation, check:

$$\begin{aligned} \boldsymbol{F}f \circ rev_A([a_1, \ldots, a_n]) &= [f(a_n), \ldots, f(a_1)] \\ &= rev_B \circ \boldsymbol{F}f([a_1, \ldots, a_n]) \end{aligned}$$

Exercise 4.2.3. Suppose that $\boldsymbol{F}, \boldsymbol{G} : \mathcal{D} \longrightarrow \mathcal{E}$, $\boldsymbol{I} : \mathcal{C} \longrightarrow \mathcal{D}$, $\boldsymbol{J} : \mathcal{E} \longrightarrow \mathcal{F}$ are functors and $\alpha : \boldsymbol{F} \stackrel{\cdot}{\longrightarrow} \boldsymbol{G}$ is a natural transformation. Verify that:

1. there is a natural transformation $\alpha_{\boldsymbol{I}} : \boldsymbol{FI} \stackrel{\cdot}{\longrightarrow} \boldsymbol{GI}$ with components $(\alpha_{\boldsymbol{I}})_C \stackrel{def}{\equiv} \alpha_{\boldsymbol{I}C}$ where $C \in ob(\mathcal{C})$.

2. there is a natural transformation $\boldsymbol{J}\alpha : \boldsymbol{JF} \stackrel{\cdot}{\longrightarrow} \boldsymbol{JG}$ with components $(\boldsymbol{J}\alpha)_D \stackrel{def}{\equiv} \boldsymbol{J}(\alpha_D)$ where $D \in ob(\mathcal{D})$.

Exercise 4.2.4. Suppose that $\mathcal{C}$ is Cartesian closed and A is an object of $\mathcal{C}$. Verify that $(\_^A) \times A \stackrel{\cdot}{\longrightarrow} \boldsymbol{id}_{\mathcal{C}}$ is a natural transformation with components $ev_B : B^A \times A \longrightarrow B$.

Definition 4.2.5 (Natural isomorphism)**.** A natural transformation with every component iso is called a *natural isomorphism*. If there is a natural isomorphism between the functors F and G, then F and G are *naturally isomorphic*.

Example 4.2.6. If a category $\mathcal{A}$ has binary products, then $\_ \times 1 \longrightarrow \boldsymbol{id}_{\mathcal{A}}$ is a natural isomorphism.

4.3 Functor Categories

Definition 4.3.1 (Functor category)**.** A *functor category* of $\mathcal{C}$ and $\mathcal{D}$ (notation $[\mathcal{C}, \mathcal{D}]$ or $\mathcal{D}^{\mathcal{C}}$) is a category with objects functors from $\mathcal{C}$ to $\mathcal{D}$, morphisms natural transformations between such functors.

Exercise 4.3.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $\boldsymbol{F}$,$\boldsymbol{G}$,$\boldsymbol{H}$ be functors from $\mathcal{C}$ to $\mathcal{D}$, and $\alpha : \boldsymbol{F} \stackrel{\cdot}{\longrightarrow} \boldsymbol{G}$ and $\beta : \boldsymbol{G} \stackrel{\cdot}{\longrightarrow} \boldsymbol{H}$ be natural transformations.

We can define a natural transformation $\beta \circ \alpha : \boldsymbol{F} \stackrel{\cdot}{\longrightarrow} \boldsymbol{H}$ by setting its components to be $(\beta \circ \alpha)_A \stackrel{def}{\equiv} \beta_A \circ \alpha_A$. Verify the following:

1. the natural transformation defined above is indeed a natural transformation.

2. functor categories are indeed categories.

Exercise 4.3.3. Show that a natural isomorphism is an isomorphism in a functor category, by the following steps:

1. Given a natural isomorphism α in $\mathcal{D}^{\mathcal{C}}$ with inverse β, show that each β_C is an inverse for α_C in $\mathcal{D}$.

2. Given a natural transformation α in $\mathcal{D}^{\mathcal{C}}$ for which each component α_C has an inverse (say β_C) in $\mathcal{D}$, show that the β_C are the components of a natural transformation β which is the inverse of α in $\mathcal{D}^{\mathcal{C}}$.

Definition 4.3.4 (Product of categories)**.** The *product* of $\mathcal{C}$ and $\mathcal{D}$ (notation $\mathcal{C} \times \mathcal{D}$) has objects pairs (C, D) where $C \in ob(\mathcal{C})$ and $D \in ob(\mathcal{D})$. The morphisms $(C, D) \longrightarrow (C', D')$ are pairs of morphisms (f, g) where $f : C \longrightarrow C'$ and $g : D \longrightarrow D'$.

Exercise 4.3.5. We can define a functor $Ev : \mathcal{S}et^{\mathcal{C}} \times \mathcal{C} \longrightarrow \mathcal{S}et$, which is defined as follows:

$$\begin{aligned} Ev(\boldsymbol{F}, \mathcal{A}) &\stackrel{def}{\equiv} \boldsymbol{F}\mathcal{A} \\ Ev(\mu, g) &\stackrel{def}{\equiv} \mu_{A'} \circ \boldsymbol{F}g = \boldsymbol{F'}g \circ \mu_A \end{aligned}$$

where $g : A \longrightarrow A'$ and $\mu : \boldsymbol{F} \stackrel{\cdot}{\longrightarrow} \boldsymbol{F'}$. Show that $\mathcal{C}at$ is Cartesian closed with functor categories as exponentials.

Definition 4.3.6 (Equivalence)**.** Two categories $\mathcal{C}$ and $\mathcal{D}$ are *equivalent* (notation $\boldsymbol{F} : \mathcal{C} \simeq \mathcal{D} : \boldsymbol{G}$) if there are functors $\boldsymbol{F} : \mathcal{C} \longrightarrow \mathcal{D}$ and $\boldsymbol{G} : \mathcal{D} \longrightarrow \mathcal{C}$ together with natural isomorphisms $\varepsilon : \boldsymbol{FG} \cong id_{\mathcal{D}}$ and $\eta : id_{\mathcal{C}} \cong \boldsymbol{GF}$. We call such $\boldsymbol{F}$ and $\boldsymbol{G}$ *functor isomorphisms.*

4.4 Covariant and Contravariant Functors

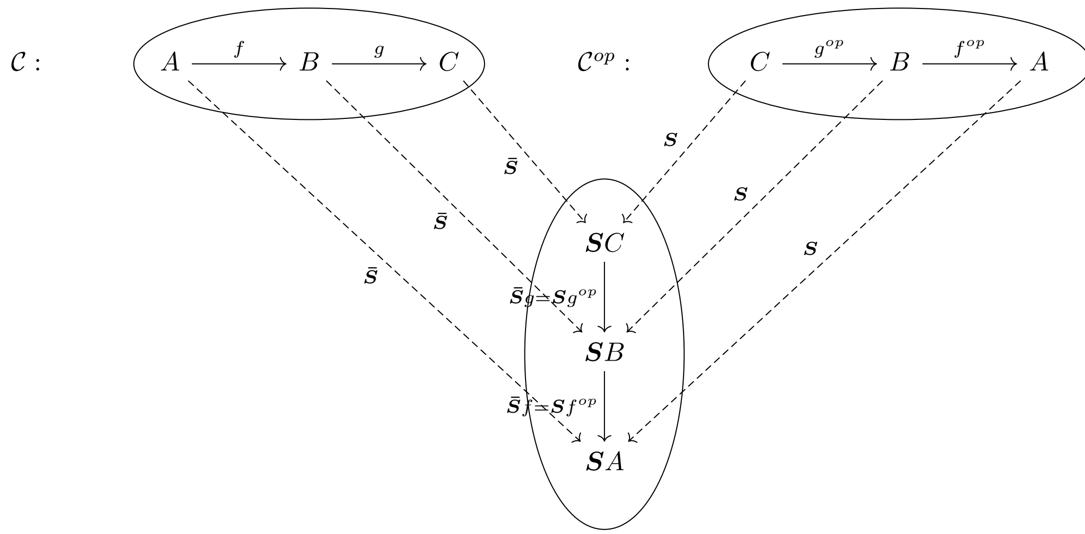
Definition 4.4.1 (Dual functor)**.** If $\boldsymbol{T} : \mathcal{C} \longrightarrow \mathcal{B}$ is a functor, then the dual of $\boldsymbol{T}$, written as $\boldsymbol{T}^{op} : \mathcal{C}^{op} \longrightarrow \mathcal{B}^{op}$ is also a functor, its object function $c \mapsto \boldsymbol{T}c$ and its mapping function $f^{op} \mapsto (\boldsymbol{T}f)^{op}$.

$$
\begin{tikzcd}
\mathcal{C}: & A \arrow[r, "f"] \arrow[d, dashed, "\boldsymbol{T}"'] & B \arrow[r, "g"] \arrow[d, dashed, "\boldsymbol{T}"'] & C \arrow[d, dashed, "\boldsymbol{T}"'] \\
\mathcal{B}: & \boldsymbol{T}A \arrow[r, "\boldsymbol{T}f"] & \boldsymbol{T}B \arrow[r, "\boldsymbol{T}g"] & \boldsymbol{T}C
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{C}^{op}: & C \arrow[r, "g^{op}"] \arrow[d, dashed, "\boldsymbol{T}^{op}"'] & B \arrow[r, "f^{op}"] \arrow[d, dashed, "\boldsymbol{T}^{op}"'] & A \arrow[d, dashed, "\boldsymbol{T}^{op}"'] \\
\mathcal{B}^{op}: & \boldsymbol{T}^{op}C \arrow[r, "\boldsymbol{T}^{op}g^{op}"] \arrow[d, equal] & \boldsymbol{T}^{op}B \arrow[r, "\boldsymbol{T}^{op}f^{op}"] \arrow[d, equal] & \boldsymbol{T}^{op}A \arrow[d, equal] \\
 & (\boldsymbol{T}C)^{op} \arrow[r, "(\boldsymbol{T}g)^{op}"] & (\boldsymbol{T}B)^{op} \arrow[r, "(\boldsymbol{T}f)^{op}"] & (\boldsymbol{T}A)^{op}
\end{tikzcd}
$$

Consider a functor $\boldsymbol{S} : \mathcal{C}^{op} \longrightarrow \mathcal{B}$ with the object function $A \mapsto \boldsymbol{S}A$ and the mapping function $f^{op} \mapsto \boldsymbol{S}f^{op}$. This functor $\boldsymbol{S}$ may be expressed in terms of the original category $\mathcal{C}$ in the following way.

Definition 4.4.2 (Contravariant functor)**.** The *contravariant functor* $\bar{\boldsymbol{S}} : \mathcal{C} \longrightarrow \mathcal{B}$ of a functor $\boldsymbol{S} : \mathcal{C}^{op} \longrightarrow \mathcal{B}$ assigns each object A in $\mathcal{C}$ an object $\boldsymbol{S}A$ in $\mathcal{B}$ and each morphism $f : A \longrightarrow B$ in $\mathcal{C}$ a morphism $\boldsymbol{S}f^{op} : \boldsymbol{S}B \longrightarrow \boldsymbol{S}A$ in $\mathcal{B}$, in such a way that the following equations hold for any object A and any morphism f, g in $\mathcal{C}$.

$$
\begin{aligned}
\bar{\boldsymbol{S}}(id_A) &= id_{\bar{\boldsymbol{S}}A} \\
\bar{\boldsymbol{S}}(g \circ f) &= \bar{\boldsymbol{S}}f \circ \bar{\boldsymbol{S}}g
\end{aligned}
$$



Remark 4.4.3. An ordinary functor defined as in Definition 4.1.1 is called a *covariant functor* in contradistinction to contravariant functors. It is often more convenient to represent a contravariant functor $\bar{\boldsymbol{S}}$ on $\mathcal{C}$ to $\mathcal{B}$ as a covariant functor $\boldsymbol{S} : \mathcal{C}^{op} \longrightarrow \mathcal{B}$ or as a covariant functor $\boldsymbol{S}^{op} : \mathcal{C} \longrightarrow \mathcal{B}^{op}$.

Example 4.4.4. A *covariant powerset functor* $\boldsymbol{P} : \mathcal{S}et \longrightarrow \mathcal{S}et$ is given by

$$f : A \longrightarrow B \quad \mapsto \quad f_* : \boldsymbol{P}A \longrightarrow \boldsymbol{P}B$$

where $f : A \longrightarrow B$ is a function and f_* is defined by $f_*(A') \overset{def}{\equiv} \{f(a') \mid a' \in A'\}$ where $A' \in \boldsymbol{P}(A)$.

Example 4.4.5. A *contravariant powerset functor* $\boldsymbol{P} : \mathcal{S}^{op} \longrightarrow \mathcal{S}et$ is given by

$$f : A \longrightarrow B \quad \mapsto \quad f^{-1} : \boldsymbol{P}(B) \longrightarrow \boldsymbol{P}(A)$$

where $f : A \longrightarrow B$ is a function in $\mathcal{S}et$, and the function f^{-1} is defined by $f^{-1}(B') \overset{def}{\equiv} \{a \in A \mid f(a) \in B'\}$ where $B' \in \boldsymbol{P}(B)$.

4.5 Full and Faithful Functors

Definition 4.5.1 (Full functor)**.** A functor $\boldsymbol{F} : \mathcal{A} \longrightarrow \mathcal{B}$ is *full* iff, for any objects A, B of $\mathcal{A}$ and any morphism $g : \boldsymbol{F}A \longrightarrow \boldsymbol{F}B$ of $\mathcal{B}$, there is some $f : A \longrightarrow B$ in $\mathcal{A}$ such that $\boldsymbol{F}f = g$.

Definition 4.5.2 (Faithful functor)**.** A functor $\boldsymbol{F} : \mathcal{A} \longrightarrow \mathcal{B}$ is *faithful* iff, for any objects A, B and any morphisms $f, g : A \longrightarrow B$ of $\mathcal{A}$, if $\boldsymbol{F}f = \boldsymbol{F}g$ then $f = g$.

Remark 4.5.3. Clearly the composite of two full functors is a full functor and the composite of two faithful functors is a faithful functor.

Example 4.5.4. Given a pair of objects $A, B \in \mathcal{C}$, the arrow function of $\boldsymbol{T} : \mathcal{C} \longrightarrow \mathcal{B}$ sends each arrow $f : A \longrightarrow B$ to $\boldsymbol{T}f : \boldsymbol{T}A \longrightarrow \boldsymbol{T}B$, so it defines a function:

$$\boldsymbol{T}_{A,B} : \mathcal{C}(A, B) \longrightarrow \mathcal{B}(\boldsymbol{T}A, \boldsymbol{T}B),\ f \mapsto \boldsymbol{T}f$$

Then $\boldsymbol{T}$ is full when every such function is surjective, and faithful when every such function is injective.

Remark 4.5.5. For a functor both full and faithful, every such function is a bijection. But this does not mean that the functor itself is an isomorphism of categories, for there may be objects of $\mathcal{B}$ not in the image of $\boldsymbol{T}$.[1]

[1] See MacLane (1997, p.15)

Bibliography

Crole, R. L. (1993) *Categories for Types.* Cambridge University Press.

Kleisli, H. (1965) “Every standard construction is induced by a pair of adjoint functors”, In the Proceedings of *the American Mathematical Society*, Vol. 16. pp.544–546.

MacLane, S. (1997) *Categories for the Working Mathematician*, Graduate Texts in Mathematics. Springer, second edition edition.

MacLane, S. and I. Moerdijk. (1992) *Sheaves in Geometry and Logic – A First Introduction to Topos Theory –*. Springer.

McLarty, C. (1992) *Elementary Categories, Elementary Toposes*, Oxford Logic Guides 21. Oxford Science Publications.

Moggi, E. (1989) “Computational lambda-calculus and monads”, In the Proceedings of *Fourth Annual IEEE Symposium on Logic in Computer Science.* pp.14–23.

Pedicchio, M. C. and W. Tholen. (2004) *Categorical Foundations: Special Topics in Order, Topology, Algebra, and Sheaf Theory*, Vol. 97 of *Encyclopedia of Mathematics and Its Applications.* Cambridge University Press.